

HW 2 – Chi-Square Test for Homogeneity

1) A study at a university wanted to see if there was a gender difference with respect to drinking behavior. Two independent random samples of male and female college students at the university asked them to rate their drinking behavior as none, low, moderate, or high. The results are shown in the table below.

	Drinking Level			
	None	Low	Moderate	High
Male	140	478	300	63
Female	180	580	285	40

a) What would be the null and alternative hypotheses to test to see if there was a gender difference with respect to drinking behavior?

b) What would the expected counts be for a male with a moderate drinking level? Show your work!

c) You run the test and get a chi-square statistic of 15.157. What is the p-value?

d) What can you conclude at the 5% significance level?

_____ 2) A survey was conducted among 340 high school girls. They were asked whether they were right-handed or left-handed and whether or not they played a musical instrument. The table below displays the results of the survey:

	Play	Does not play	Total
Right-Handed	74	236	310
Left-Handed	18	12	30
Total	92	248	340

If the likelihood of a girl playing an instrument is independent of left or right-handedness, what is the expected count representing right-handed girls who play a musical instrument?

- (A) 62
- (B) 74
- (C) 80.2
- (D) 83.9
- (E) 92.1

_____ 3) Each person in a random sample of university students was asked which of five flavors of ice cream he or she preferred. The resulting data was used to test the null hypothesis $H_o: p_1 = p_2 = \dots = p_5 = 0.2$. The P-value associated with this test was 0.067. If a significance level of 0.05 is used for the test, which of the following is an appropriate conclusion?

- (A) Fail to reject H_o , there is convincing evidence that there is no preference for one flavor of ice cream over another.
- (B) Fail to reject H_o , there is not convincing evidence that there is a preference for one flavor of ice cream over another.
- (C) Reject H_o , there is convincing evidence that there is no preference for one flavor of ice cream over another.
- (D) Reject H_o , there is convincing evidence that there is a preference for one flavor of ice cream over another.
- (E) Based on the information given, it is not possible to determine whether H_o should be rejected or not.

_____ 4) A geneticist claims that four species of fruit flies should appear in the ratio 1:3:3:9. Suppose that a sample of 480 flies contained 25, 92, 68, and 295 flies of each species, respectively. Does a chi-square test show sufficient evidence to reject the geneticist's claim at the 1% significance level?

- (A) The test proves the geneticist's claim.
- (B) The test proves the geneticist's claim is false.
- (C) The test does not give sufficient evidence to reject the geneticist's claim.
- (D) The test gives sufficient evidence to reject the geneticist's claim.
- (E) The test is inconclusive.